

NONCENTRAL MULTIVARIATE DIRICHLET DISTRIBUTIONS

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1. Introduction and Summary

Suppose that x_j ($j = 1, \dots, k$) and y are independent random variables having chi-square distributions with m_j ($j = 1, \dots, k$) and n degrees of freedom respectively. The joint distribution of $(z_1, \dots, z_k)$, where $z_j = x_j / (\sum_1^k x_j + y)$, $j = 1, \dots, k$ is the Dirichlet distribution $D(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$ with density function given by (Wilks (1962)).

$$(1) \quad \Gamma[\frac{1}{2}(m+n)] / \{\prod_1^k \Gamma(\frac{1}{2}m_j) \Gamma(\frac{1}{2}n)\} \prod_1^k z_j^{\frac{1}{2}m_j-1} (1-\sum_1^k z_j)^{\frac{1}{2}n-1}$$

where

$$z_j > 0, \quad 1-\sum_1^k z_j > 0 \quad \text{and} \quad m = \sum_1^k m_j.$$

If v_j , $j=1, \dots, k$ is defined by $v_j = x_j/y$, $j=1, \dots, k$ then the joint density of $(v_1, \dots, v_k)$ is

$$(2) \quad \Gamma[\frac{1}{2}(m+n)] / \{\prod_1^k \Gamma(\frac{1}{2}m_j) \Gamma(\frac{1}{2}n)\} \prod_1^k v_j^{\frac{1}{2}m_j-1} \cdot (1+\sum_1^k v_j)^{-\frac{1}{2}(m+n)},$$

$$v_j > 0$$

with $m = \sum_1^k m_j$.

This distribution could also be called a Dirichlet distribution. Because (1) reduces to the Beta Type 1 distribution and (2) reduces to the Beta Type 2 distribution we shall call the first a Dirichlet Type 1 distribution $D_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$ and the second a Dirichlet Type 2 distribution $D_2(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$.

If y has a noncentral chi-square distribution (noncentrality parameter κ) then we have noncentral Dirichlet distributions. The joint distribution of $(z_1, \dots, z_k)$ is then the noncentral

Dirichlet Type 1 distribution $D_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \kappa)$ with density function

$$(3) \quad \frac{\Gamma[\frac{1}{2}(m+n)]}{\Gamma(\frac{1}{2}n) \prod_1^k \Gamma(\frac{1}{2}m_j)} \prod_1^k z_j^{\frac{1}{2}m_j-1} (1-\sum_1^k z_j)^{\frac{1}{2}n-1}$$

$$\cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\kappa^2 (1-\sum_1^k z_j)],$$

$$z_j > 0, \quad 1-\sum_1^k z_j > 0,$$

where $m = \sum_1^k m_j$.

Similarly the joint distribution of $(v_1, \dots, v_k)$ is the noncentral Dirichlet Type 2 distribution $D_2(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \kappa)$ with the density function

$$(4) \quad \frac{\Gamma[\frac{1}{2}(m+n)]}{\Gamma(\frac{1}{2}n)\prod_1^k \Gamma(\frac{1}{2}m_j)} \prod_1^k v_j^{\frac{1}{2}m_j-1} (1+\sum_1^k v_j)^{-\frac{1}{2}(m+n)} \\ \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\kappa^2(1+\sum_1^k v_j)^{-1}], \quad v_j > 0,$$

with $m = \sum_1^k m_j$.

These distributions can be derived as special cases from the noncentral multivariate Dirichlet distributions which will be considered in 2. In 3. the moments of the Dirichlet determinants will be derived. The complex multivariate Dirichlet distributions will be considered in 4.

The moments of the real and complex Dirichlet determinants are of considerable interest in multivariate likelihood ratio tests. In 5 it will be shown how these moments can be used to find the moments of the criterion for testing the identity of several multivariate normal distributions.

2. Noncentral Multivariate Dirichlet Distributions

Let $A_j : p \times p$ ($j = 1, \dots, k$) and $B : p \times p$ be independent random matrices. Suppose that A_j ($j = 1, \dots, k$) has a Wishart distribution with m_j degrees of freedom and that B has a noncentral Wishart distribution with n degrees of freedom and noncentrality parameter κ (only the linear case is treated).

Let L_j be defined by

$$(5) \quad L_j = C^{-1} A_j C'^{-1}, \quad j = 1, \dots, k$$

where

$$(6) \quad \sum_1^k A_j + B = CC'$$

and C is a lower triangular matrix.

The joint distribution of $L_1, \dots, L_k$ is the noncentral multivariate Dirichlet Type 1 distribution $MD_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \kappa)$ with density function (Troskie (1966)).

$$(7) \quad \frac{\Gamma_p[\frac{1}{2}(m+n)]}{\Gamma_p(\frac{1}{2}n)\prod_1^k \Gamma_p(\frac{1}{2}m_j)} \prod_1^k |L_j|^{\frac{1}{2}(m_j-p-1)} |I - \sum_1^k L_j|^{\frac{1}{2}(n-p-1)} \\ \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\kappa^2(1 - \sum_1^k L_{j11})],$$

$$L_j > 0, \quad I - \sum_1^k L_j > 0,$$

where $m = \sum_1^k m_j$, $A > 0$ means that A is positive definite,

$$L_j = [L_{j\alpha\beta}]; \quad j = 1, \dots, k; \quad \alpha, \beta = 1, \dots, p$$

and

$$(8) \quad \Gamma_p(\delta) = \pi^{\frac{1}{2}p(p-1)} \Gamma(\delta) \Gamma(\delta - \frac{1}{2}) \dots \Gamma(\delta - \frac{1}{2}(p-1))$$

is the multivariate gamma function.

If $V_j (j = 1, \dots, k)$ is defined by

$$(9) \quad V_j = B^{-\frac{1}{2}} A_j B^{-\frac{1}{2}}, \quad j = 1, \dots, k$$

then the joint distribution of $V_1, \dots, V_k$ is the noncentral multivariate Dirichlet Type 2 distribution $MD_2(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \kappa)$ with density function (Troskie (1966)).

$$(10) \quad \frac{\Gamma_p[\frac{1}{2}(m+n)]}{\Gamma_p(\frac{1}{2}n) \prod_1^k \Gamma_p(\frac{1}{2}m_j)} \pi_1^k |V_j|^{\frac{1}{2}(m_j-p-1)} |I + \sum_1^k V_j|^{-\frac{1}{2}(m+n)} \\ \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\kappa^2 \lambda^{11}], \quad V_j > 0,$$

with $m = \sum_1^k m_j$, $\Lambda = I + \sum_1^k V_j$ and

$$\Lambda^{-1} = [\lambda^{ij}], \quad i, j = 1, 2, \dots, p.$$

If $\kappa = 0$ B has a central Wishart distribution. The distributions (7) and (10) then reduces to the central multivariate Dirichlet distributions $MD_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$ and $MD_2(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$, respectively. These distributions have already been investigated by Olkin & Rubin (1964).

For $p = 1$ the distributions (7) and (10) reduces to the noncentral Dirichlet distributions (3) and (4) respectively.

3. The Moments Of The Dirichlet Determinants

The moments of the determinants $|L_j|$ and $|V_j|$, $j = 1, \dots, k$ are of considerable interest in multivariate likelihood ratio tests (for example testing that k multivariate normal populations are identical; see 5.).

To obtain the moments of $|L_j|$ and $|V_j|$ one needs the following:

Lemma 3.1

If L_j , $j = 1, \dots, k$ is partitioned as

$$(11) \quad L_j = \begin{pmatrix} L_{j11} & L_{j12} \\ L_{j21} & L_{j22} \end{pmatrix}; \quad j = 1, \dots, k,$$

where L_{j11} is a $(q \times q)$ matrix, then the marginal distribution of $L_{111}, L_{211}, \dots, L_{k11}$ has a Dirichlet distribution $MD_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$ with density function

$$(12) \quad \frac{\Gamma_q[\frac{1}{2}(m+n)]}{\Gamma_q(\frac{1}{2}n) \prod_1^k \Gamma_q(\frac{1}{2}m_j)} \pi_1^k |L_{j11}|^{\frac{1}{2}(m_j-q-1)} |I_q - \sum_1^k L_{j11}|^{\frac{1}{2}(n-q-1)}$$

$$L_{j11} > 0, \quad I_q - \sum_1^k L_{j11} > 0$$

and $m = \sum_1^k m_j$.

Proof

Let $A_j = \begin{pmatrix} A_{j11} & A_{j12} \\ A_{j21} & A_{j22} \end{pmatrix}$, $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$ and $C = \begin{pmatrix} C_{11} & 0 \\ C_{21} & C_{22} \end{pmatrix}$, where A_{j11} and B_{11} are $(q \times q)$ matrices and C_{11} is a $(q \times q)$ lower triangular matrix. Since $\Sigma_1^k A_j + B = CC'$ and $A_j = CL_j C'$, $j=1, \dots, k$, it follows that $\Sigma_1^k A_{11} + B_{11} = C_{11} C'_{11}$ and $A_{j11} = C_{11} L_{j11} C'_{11}$. But A_{j11} , $j = 1, \dots, k$, and B_{11} have independent Wishart distributions, therefore $L_{111}, \dots, L_{k11}$, is distributed according to $MD_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$.

Theorem 3.1

If $L_1, \dots, L_k$ is distributed according to $MD_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \kappa)$, then

$$(13) \quad E[\pi_1^k | L_j |^{r_j}] = \frac{\Gamma_p[\frac{1}{2}(m+n)] \pi_1^k \Gamma_p(\frac{1}{2}m_j + r_j)}{\pi_1^k \Gamma_p(\frac{1}{2}m_j) \Gamma_p[\frac{1}{2}(m+n) + r]} \\ \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}(m+n) + r; \frac{1}{2}\kappa^2],$$

with $m = \Sigma_1^k m_j$, $r = \Sigma_1^k r_j$, and $r_j \geq -\frac{1}{2}(m_j - p)$.

Proof

$$E[\pi_1^k | L_j |^{r_j}] = \frac{\Gamma_p[\frac{1}{2}(m+n)]}{\Gamma_p(\frac{1}{2}n) \pi_1^k \Gamma_p(\frac{1}{2}m_j)} \int \dots \int \pi_1^k |L_j|^{\frac{1}{2}(m_j + 2r_j - p - 1)} \\ \cdot |I - \Sigma_1^k L_j|^{\frac{1}{2}(n-p-1)} e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\kappa^2 (1 - \Sigma_1^k L_{j11})] \\ \cdot dL_1 \dots dL_k \\ = \frac{\Gamma_p[\frac{1}{2}(m+n)]}{\pi_1^k \Gamma_p(\frac{1}{2}m_j) \Gamma_p(\frac{1}{2}n)} \cdot \frac{\Gamma_p(\frac{1}{2}n) \pi_1^k \Gamma_p(\frac{1}{2}m_j + r_j)}{\Gamma_p[\frac{1}{2}(m+n) + r]} \\ \cdot \int \dots \int \left[\frac{\Gamma_p[\frac{1}{2}(m+n) + r]}{\Gamma_p(\frac{1}{2}n) \pi_1^k \Gamma_p(\frac{1}{2}m_j + r_j)} \cdot \pi_1^k |L_j|^{\frac{1}{2}(m_j + 2r_j - p - 1)} \cdot |I - \Sigma_1^k L_j|^{\frac{1}{2}(n-p-1)} \right] \\ \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\kappa^2 (1 - \Sigma_1^k L_{j11})] dL_1 \dots dL_k.$$

The term within the brackets is the density function of $MD_1(\frac{1}{2}m_1 + r_1, \dots, \frac{1}{2}m_k + r_k; \frac{1}{2}n)$. Integrating this density function

over all elements of L_j ($j=1, \dots, k$) except the first element of each L_j , one obtains the marginal density of z_{j11} ($j=1, \dots, k$) which is distributed as $D_1(\frac{1}{2}m_1+r_1, \dots, \frac{1}{2}m_k+r_k; \frac{1}{2}n)(q=1)$.

Therefore

$$\begin{aligned} E[\pi_1^k | L_j]^{r_j} &= \frac{\Gamma_p[\frac{1}{2}(m+n)] \pi_1^k \Gamma_p(\frac{1}{2}m_j+r_j)}{\pi_1^k \Gamma_p(\frac{1}{2}m_j) \Gamma_p[\frac{1}{2}(m+n)+r]} \\ &\cdot \dots \int \frac{\Gamma[\frac{1}{2}(m+n)+r]}{\Gamma(\frac{1}{2}n) \pi_1^k \Gamma(\frac{1}{2}m_j+r_j)} \pi_1^k z_{j11}^{\frac{1}{2}m_j+r_j-1} (1-\Sigma_1^k z_{j11})^{\frac{1}{2}n-1} \\ &\cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\kappa^2(1-\Sigma_1^k z_{j11})] d z_{111} \dots d z_{k11}. \end{aligned}$$

The results follows since

$$\begin{aligned} &\int \dots \int \pi_1^k z_{j11}^{\frac{1}{2}m_j+r_j-1} (1-\Sigma_1^k z_{j11})^{\frac{1}{2}n-1} \\ &\cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\kappa^2(1-\Sigma_1^k z_{j11})] d z_{111}, \dots, d z_{k11} \\ &= \frac{\Gamma(\frac{1}{2}n) \pi_1^k \Gamma(\frac{1}{2}m_j+r_j)}{\Gamma[\frac{1}{2}(m+n)+r]} \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}(m+n)+r; \frac{1}{2}\kappa^2]. \end{aligned}$$

Theorem 3.2

If $V_1, \dots, V_k$ is distributed as $MD_2(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \kappa)$ then

$$(14) \quad E[\pi_1^k | V_j]^{r_j} = \frac{\pi_1^k \Gamma_p(\frac{1}{2}m_j+r_j) \Gamma_p(\frac{1}{2}n-r)}{\pi_1^k \Gamma_p(\frac{1}{2}m_j) \Gamma_p(\frac{1}{2}n)} \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1(\frac{1}{2}n-r; \frac{1}{2}n; \frac{1}{2}\kappa^2)$$

with $m = \Sigma_1^k m_j$ and $r = \Sigma_1^k r_j$.

Proof

since

$$E[\pi_1^k | V_j]^{r_j} = E[\pi_1^k | L_j]^{r_j} | I - \Sigma_1^k L_j |^{-r}$$

this expectation is proportional to the expectation of $\pi_1^k | L_j]^{r_j}$ when the joint distribution function of the L_j is $MD_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}(n-2r); \kappa)$. Evaluation of the constant of proportionality together with Theorem 3.1 gives the result.

4. Complex Multivariate Dirichlet Distributions

Let $Z : p \times n$ be a complex random matrix; that is $Z = X + iY$ where X and Y are real matrices. If Z has a complex multivariate normal distribution $CN(\mu, \Sigma)$ then the density function of Z is (Goodman (1963), Khatri (1965))

$$(15) \pi^{-pn} |\Sigma|^{-n} \text{etr}[-\Sigma^{-1}(Z-\mu)(\overline{Z-\mu})^t]$$

where $\text{etr} = \exp(\text{tr})$, Σ is Hermitian positive definite (hpd) and $\mu : p \times n$ is of rank t ($t \leq p \leq n$).

Let

$$(16) Z\overline{Z}^t = A$$

be the complex Wishart matrix. The distribution of A can be explicitly given for the special cases when $t = 0$ and $t = 1$. For $t = 0$ the distribution of A is the central complex Wishart distribution with density function (Goodman (1963), Khatri (1965)),

$$(17) CW(A, \Sigma, n) = [\bar{\Gamma}_p(n) |\Sigma|^n]^{-1} |A|^{n-p} \text{etr}(-\Sigma^{-1} A)$$

where

$$\bar{\Gamma}_p(n) = \pi^{\frac{1}{2}p(p-1)} \pi_1^p \Gamma(n-j+1) \text{ and } A \text{ is hpd.}$$

For $t = 1$ the distribution of A is the noncentral complex Wishart distribution (linear case) with density function (Khatri (1965))

$$(18) CW(A, \Sigma, n) \exp(-\alpha) {}_0F_1(n; \alpha_1)$$

where $\alpha_1 = \text{tr}(\bar{\mu}^t \Sigma^{-1} A \Sigma^{-1} \mu)$, $\alpha = \text{tr}(\bar{\mu}^t \Sigma^{-1} \mu)$ and A is hpd.

From the complex Wishart distribution $CW(\Sigma, n)$ one can derive the complex multivariate Dirichlet distributions. For the derivation the following Jacobians are necessary.

Let A be hpd and let T be an upper triangular matrix with t_{jj} real, > 0 , $j = 1, \dots, p$ and $t_{jk} = t_{jkR} + it_{jkI}$. The Jacobian of the transformation $A = \bar{T}^t T$ is (Goodman (1963))

$$(19) J(A \rightarrow T) = 2^p \pi_1^p \prod_{j=1}^p t_{jj}^{2(p-j)+1}.$$

Let A and B be hpd matrices and let $B^{\frac{1}{2}}$ be the hpd square root of B . The Jacobian of the transformation

$$A = B^{\frac{1}{2}} V B^{\frac{1}{2}} \text{ is (Khatri (1965))}$$

$$(20) J(A \rightarrow V) = |B|^p.$$

Lemma 4.1

Let A be hpd and let T be an upper triangular matrix defined as above. The Jacobian of the transformation

$$A = \bar{T}^t L T \text{ is}$$

$$(21) J(A \rightarrow L) = |\bar{T}^t T|^p.$$

Proof

Since T is an upper triangular matrix

$$a_{ij} = \sum_{k, l} \bar{t}_{ki} \bar{t}_{kl} t_{lj}$$

with $t_{ij} = 0$ for $i > j$. Writing the Jacobian in the following form

$$J = \frac{\partial(a_{11}, a_{12R}, a_{12I}, \dots, a_{1pR}, a_{1pI}, a_{22}, \dots, a_{pp})}{\partial(\bar{t}_{11}, \bar{t}_{12R}, \bar{t}_{12I}, \dots, \bar{t}_{1pR}, \bar{t}_{1pI}, \bar{t}_{22}, \dots, \bar{t}_{pp})}$$

one immediately sees that the Jacobian matrix is triangular with the diagonal elements being given by

$$\frac{\partial a_{jj}}{\partial t_{jj}} = t_{jj}, \quad \frac{\partial a_{jkR}}{\partial t_{jkR}} = t_{jj} t_{kk},$$

$$\frac{\partial a_{jkI}}{\partial t_{jkI}} = t_{jj} t_{kk}; \quad j < k, \quad j, k=1, \dots, p.$$

Thus

$$\begin{aligned} J(A \rightarrow L) &= (t_{11}^2)^p (t_{22}^2)^p \dots (t_{pp}^2)^p \\ &= |T|^{2p} = |\bar{T}' T|^p. \end{aligned}$$

Theorem 4.1

Let A_j , $j=1, \dots, k$ and B be independently distributed hpd random matrices with distributions $CW(\Sigma, m_j)$, $j=1, \dots, k$ and $CW(\Sigma, n)$, respectively. The joint distribution of

$$(22) \quad L_j = \bar{T}'^{-1} A_j T^{-1}, \quad j=1, \dots, k$$

where $\Sigma_j^k A_j + B = \bar{T}' T$ and T is upper triangular is the complex multivariate Dirichlet Type I distribution $CMD_1(m_1, \dots, m_k; n)$ with density function

$$(23) \quad \bar{\Gamma}_p^{(m+n)} / \{ \Pi_1^k \bar{\Gamma}_p(m_j) \bar{\Gamma}_p(n) \} \Pi_1^k |L_j|^{m_j - p} |I - \Sigma_j^k L_j|^{n-p}$$

with

$$L_j \text{ and } I - \Sigma_j^k L_j \text{ hpd and } m = \Sigma_j^k m_j.$$

Proof

In the joint distribution of A_j , $j=1, \dots, k$ and B make the transformation (22). The Jacobian is

$$J(A_1, \dots, A_k, B \rightarrow L_1, \dots, L_k, T) = 2^p \Pi_1^p t_{jj}^{2(p-j)+1} |\bar{T}' T|^{kp}.$$

The joint distribution of L_j , $j=1, \dots, k$ and T is

$$\begin{aligned} & [\bar{\Gamma}_p(n) \Pi_1^k \bar{\Gamma}_p(m_j)]^{-1} |\Sigma|^{-(m+n)} \Pi_1^k |L_j|^{m_j - p} \\ & |I - \Sigma_j^k L_j|^{n-p} |\bar{T}' T|^{(m+n-p)} \text{etr}(-\Sigma^{-1} \bar{T}' T) 2^p \Pi_1^p t_{jj}^{2(p-j)+1} \end{aligned}$$

Making the transformation $G = \bar{T}' T$ with $J(T \rightarrow G) = 2^{-p} \Pi_1^p t_{jj}^{-2(p-j)-1}$ and integrating over G (hpd) yields the result.

Theorem 4.2

Let A_j , $j=1, \dots, k$ and B be independent hpd random matrices with

distributions $CW(I, m_j)$, $j=1, \dots, k$ and $CW(I, n)$, respectively. The joint distribution of $V_1, \dots, V_k$ where

$$(24) \quad A_j = B^{\frac{1}{2}} V_j B^{\frac{1}{2}}, \quad j=1, \dots, k$$

is the complex multivariate Dirichlet Type 2 distribution $CMD_2(m_1, \dots, m_k; n)$ with density function

$$(25) \quad \bar{\Gamma}_p(m+n) / \{ \prod_1^k \bar{\Gamma}_p(m_j) \bar{\Gamma}_p(n) \} \prod_1^k |V_j|^{m_j - p} |I + \sum_1^k V_j|^{-(m+n)}$$

where

$$V_j \text{ is hpd and } m = \sum_1^k m_j.$$

Proof

In the joint distribution of $A_1, \dots, A_k$ and B make the transformation (24) the Jacobian being $J(A_1, \dots, A_k \rightarrow V_1, \dots, V_k) = |B|^{kp}$ (see (20)) and obtain the joint distribution of $V_1, \dots, V_k$ and B . Integrating over B (hpd) yields the result.

For $k = 1$ the complex multivariate Dirichlet distributions reduces to the complex multivariate Beta distributions. These distributions have been investigated by Khatri (1965).

From the noncentral complex Wishart distribution one can derive the noncentral complex multivariate Dirichlet distributions. To simplify the derivations the complex Wishart distribution will be transformed to its canonical form. Since $M = \mu \bar{\mu}'$ is of rank $t = 1$ and Σ is hpd there exist a nonsingular matrix C such that

$$C \bar{M} C' = D_{\kappa}^2 = \text{diag}(\kappa^2, 0, \dots, 0) \text{ and } C \bar{\Sigma} C' = I.$$

Let B be defined by $B = C \bar{A} C'$, then the density of B is

$$(26) \quad CW(B, I, n) \exp(-\alpha)_0 F_1(n; \alpha_1)$$

where

$$\alpha = \kappa^2 \text{ and } \alpha_1 = \kappa^2 b_{11}.$$

Theorem 4.3

Let A_j , $j=1, \dots, k$ have a central complex Wishart distribution $CW(I, m_j)$, $j=1, \dots, k$ and let B be independently distributed according to a noncentral complex Wishart distribution with density function given by (26).

The joint density of $L_1, \dots, L_k$ where L_j is defined by (22), is the noncentral complex multivariate Dirichlet Type 1 distribution $CMD_1(m_1, \dots, m_k; n; \kappa)$ with density function

$$(27) \quad \bar{\Gamma}_p(m+n) / \{ \prod_1^k \bar{\Gamma}_p(m_j) \bar{\Gamma}_p(n) \} \prod_1^k |L_j|^{m_j - p} |I - \sum_1^k L_j|^{n-p} \\ \cdot e^{-\kappa^2} {}_1F_1[m+n; n; \kappa^2 (1 - \sum_1^k L_{11})]$$

with L_j and $I - \sum_1^k L_j$ hpd and $m = \sum_1^k m_j$.

Proof

To prove the above one proceeds in exactly the same way as in Theorem 4.1 except that when integrating over G one uses the fact that

$$(28) \int g_{11}^j |G|^{m+n-p} \text{etr}(-G) dG = \bar{\Gamma}_p(m+n) \Gamma(m+n+j) / \Gamma(m+n).$$

Theorem 4.4

Under the assumptions of Theorem 4.3 the joint distribution of V_j , $j=1, \dots, k$ where V_j is defined by (24) is the noncentral complex multivariate Dirichlet Type 2 distribution $\text{CMD}_2(m_1, \dots, m_k; n; \kappa)$ with density function

$$(29) \frac{\bar{\Gamma}_p(m+n)}{(\bar{\Gamma}_p(n) \prod_1^k \bar{\Gamma}_p(m_j))} \prod_1^k |V_j|^{m_j-p} |I + \Sigma_1^k V_j|^{-(m+n)} \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1(m+n; n; \kappa^2 \omega^{11})$$

where V_j is hpd , $m = \Sigma_1^k m_j$ and

$$\Omega^{-1} = (\omega^{ij}) = (I + \Sigma_1^k V_j)^{-1}.$$

Proof

Using Theorem 4.2 and the fact that

$$\int b_{11}^j |B|^{m+n-p} \text{etr}(-\Sigma^{-1}B) dB = \bar{\Gamma}_p(m+n) |\Sigma|^{m+n} \sigma_{11}^j \Gamma(m+n+j) / \Gamma(m+n)$$

where $\Sigma^{-1} = \Omega$, the result follows immediately.

For $k=1$ the noncentral complex multivariate Dirichlet distributions reduces to the noncentral complex Beta distributions (Khatri (1965)).

To find the moments of the complex determinants $|L_j|$ and $|V_j|$ one proceeds in the same way as for real determinants. It is easy to show that if $L_1, \dots, L_k$ are distributed according to $\text{CMD}_1(m_1, \dots, m_k; n; \kappa)$ then

$$(30) E\{\prod_1^k |L_j|^{r_j}\} = \frac{\bar{\Gamma}_p(m+n) \prod_1^k \bar{\Gamma}_p(m_j+r_j)}{\prod_1^k \bar{\Gamma}_p(m_j) \bar{\Gamma}_p(m+n+r)} \cdot e^{-\kappa^2} {}_1F_1(m+n; m+n+r; \kappa^2)$$

with $m = \Sigma m_j$ and $r = \Sigma r_j$.

Similarly if $V_1, \dots, V_k$ are distributed according to $\text{CMD}_2(m_1, \dots, m_k; n; \kappa)$ then

$$(31) E\{\prod_1^k |V_j|^{r_j}\} = \frac{\bar{\Gamma}_p(n-r) \prod_1^k \bar{\Gamma}_p(m_j+r_j)}{\bar{\Gamma}_p(m+n+r) \prod_1^k \bar{\Gamma}_p(m_j)} \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1(n-r; n; \kappa^2).$$

5. Applications

Consider k multivariate normal distributions with different mean vectors and covariance matrices. A criterion for testing the hypothesis of identity of these distributions is given by (Anderson (1958))

$$(32) \quad \Lambda = \pi_1^k |A_j|^{1/2 m_j} / |A+B|^{1/2 m}$$

where $A = \sum_1^k A_j$, A_j is the "within" matrix of sums of squares and cross products of the sample of the j -th population and B is the "between" matrix of sums of squares and cross products.

Under the null hypothesis A , A_j and B ($k > p$) have central Wishart distributions with

$$m = \sum_1^k m_j, \quad m_j \quad \text{and} \quad n = k-1$$

degrees of freedom respectively.

If L_j is defined by

$$A_j = C L_j C' \quad \text{where} \quad \sum_1^k A_j + B = CC',$$

then

$$\begin{aligned} \Lambda &= \pi_1^k |A_j|^{1/2 m_j} / |A+B|^{1/2 m} \\ &= \pi_1^k |L_j|^{1/2 m_j}. \end{aligned}$$

Since $(L_1, \dots, L_k)$ is now distributed as $MD_1(1/2 m_1, \dots, 1/2 m_k; 1/2 n)$ the moments of Λ follow from

$$\begin{aligned} (33) \quad E(\Lambda^h) &= E\{\pi_1^k |L_j|^{1/2 m_j h}\} \\ &= \frac{\Gamma_p[1/2(m+n)] \pi_1^k \Gamma_p[1/2(m_j+m_j h)]}{\pi_1^k \Gamma_p(1/2 m_j) \Gamma_p[1/2(m+n+m_h)]} \end{aligned}$$

where $m = \sum_1^k m_j$. The moments of Λ were first derived by Wilks (1932a) (see Anderson (1958), page 253).

The null hypothesis can also be tested by first testing the hypothesis of the equality of covariance matrices and then the hypothesis of equality of the mean vectors (Anderson (1958)).

If the covariances are identical but the mean vectors are different, then B has a noncentral Wishart distribution with $n = k-1$ degrees of freedom. Let the matrix of noncentrality parameters be of rank 1 (linear case). The joint distribution of $L_1, \dots, L_k$ is then $MD_1(1/2 m_1, \dots, 1/2 m_k; 1/2 n; \kappa)$ and the moments of Λ follow from (13), as

$$(34) \quad E(\Lambda)^h = E\{\pi_1^k |L_j|^{1/2 m_j h}\}$$

$$= \frac{\Gamma_p[\frac{1}{2}(m+n)] \pi_1^k \Gamma_p[\frac{1}{2}(m_j+m_jh)]}{\pi_1^k \Gamma_p(\frac{1}{2}m_j) \Gamma_p[\frac{1}{2}(m+n+mh)]} \cdot e^{-\frac{1}{2}\kappa^2} {}_1F_1[\frac{1}{2}(m+n); \frac{1}{2}(m+n+mh); \frac{1}{2}\kappa^2]$$

with $m = \sum_1^k m_j$.

It is also of importance to test the identity of several complex multivariate distributions. Using recent results obtained by Giri (1965) and Khatri (1965) one can show that the criterion for testing this hypothesis is

$$\Lambda = \pi_1^k |A_j|^{m_j} / |A+B|^m \\ = \pi_1^k |L_j|^{m_j}$$

where $m = \sum_1^k m_j$ and A , A_j and B have the same meaning as before but are defined in terms of complex variates. Under the null hypothesis $(L_1, \dots, L_k)$ is distributed according to

$CMD_1(m_1, \dots, m_k; n)$ and the moments of Λ are

$$(35) \quad E(\Lambda)^h = E\{\pi_1^k |L_j|^{m_jh}\} \\ = \frac{\bar{\Gamma}_p(m+n) \pi_1^k \bar{\Gamma}_p(m_j+m_jh)}{\pi_1^k \bar{\Gamma}_p(m_j) \bar{\Gamma}_p(m+n+m_jh)}.$$

Similarly if the covariance matrices are equal but the mean vectors are different then the moments of Λ are (see (30))

$$(36) \quad E(\Lambda)^h = E\{\pi_1^k |L_j|^{m_jh}\} \\ = \frac{\bar{\Gamma}_p(m+n) \pi_1^k \bar{\Gamma}_p(m_j+m_jh)}{\pi_1^k \bar{\Gamma}_p(m_j) \bar{\Gamma}_p(m+n+m_jh)} \cdot e^{-\kappa^2} {}_1F_1(m+n; m+n+m_jh; \kappa^2)$$

The distributional results in complex multivariate analysis are applicable in describing the statistical variable of empirical spectral analysis in multiple stationary time series. The complex Wishart given by (17) describes the joint statistical variability of estimators for the elements of a spectral density matrix (Goodman (1963), Rosenblatt (1962)). The role that is played by the complex multivariate Beta and Dirichlet distributions in this field will be investigated.

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